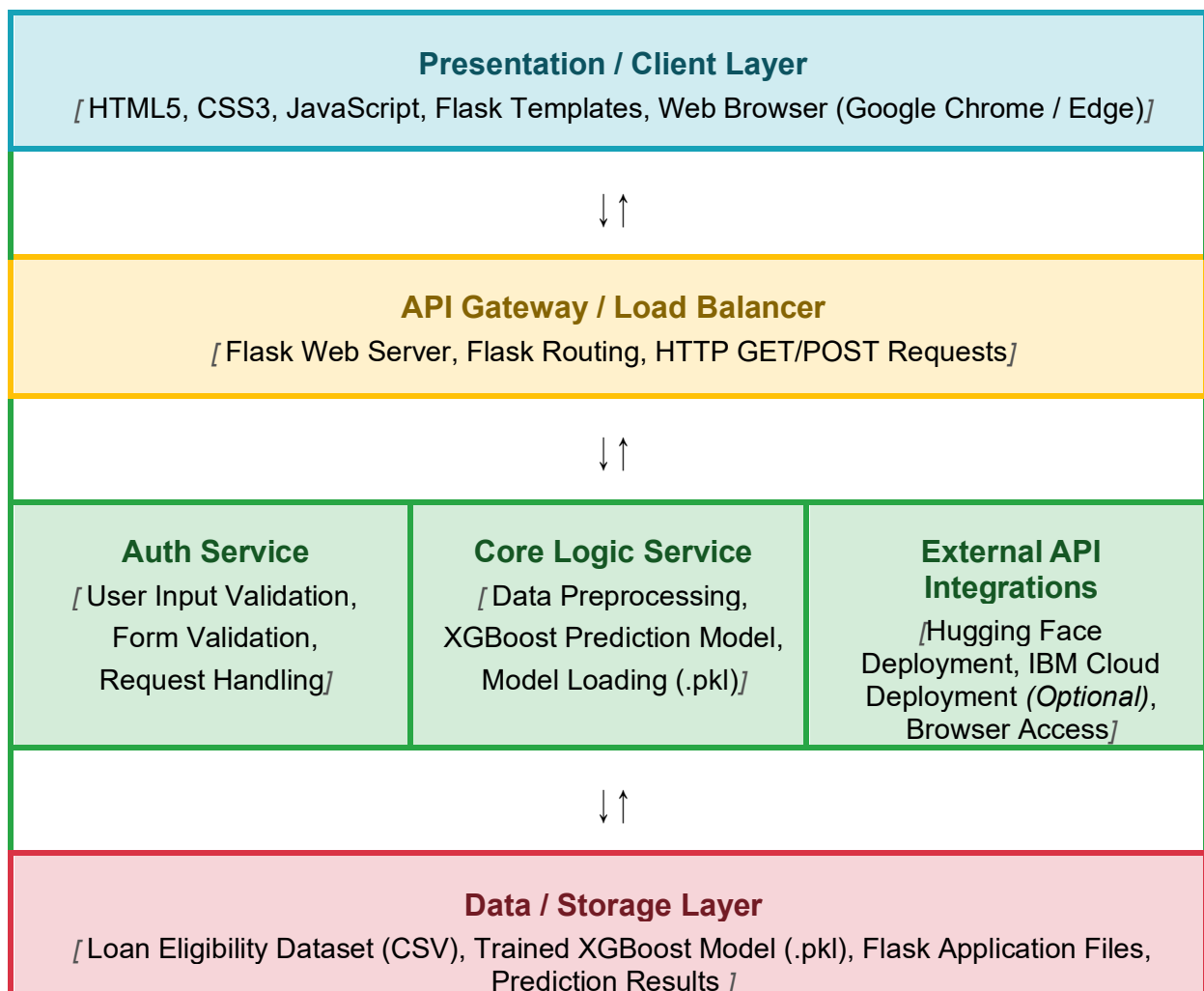


Solution Architecture

Date	2 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering applicant details and displaying loan prediction results.	HTML, CSS, JavaScript, Flask Templates
Application Layer (Flask)	Handles user requests, validates input, processes applicant data, loads the trained model, and generates loan predictions.	Python, Flask
Machine Learning Service	Performs data preprocessing and predicts loan eligibility using the trained XGBoost model.	Scikit-learn, XGBoost, Pandas, NumPy
Data / Storage Layer	Stores the loan dataset, trained model (.pkl), and application files required for prediction.	CSV Dataset, Pickle (.pkl), File System